

# The interaction of meaning similarity and confusability explains regularity in form–meaning mappings at and below the word level

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Languages exhibit striking regularities in how meanings are mapped to word forms, yet analogous patterns at the subword level remain under-explored. This study presents a large-scale cross-linguistic analysis of regularity at and below the word level, drawing on data from over 1,900 languages. Here we find that while the co-expression of meanings at both levels is highly systematic, the meanings recurrently involved in the two levels differ. Nevertheless, regularity at and below the word level is explained by the same underlying principle: a tension between pressure for lexical compression and lexical differentiation. The former favours reusing forms for similar meanings to ease learning and processing. The latter favours the use of distinct forms to prevent ambiguity. These findings offer a unified account of lexical organization across the world's languages, with subword-level form reuse emerging as a principled compromise when full word-level reuse risks miscommunication.

Linguistic regularities exist at all levels, from the sound inventories of languages to the way these sounds are combined to form words and how these words are combined to form sentences<sup>1–6</sup>. The same is true of how languages categorize and organize meaning<sup>7–10</sup>. One such cross-linguistic regularity is that related meanings are more likely to be mapped to the same form<sup>11–16</sup>. That is, many languages use the same word form for related meanings (for example, they use the same word for the meanings ‘tree’ and ‘wood’, or ‘mouth’ and ‘word’). Such co-expressions of meanings are called colexifications<sup>17,18</sup>, and they are cross-linguistically much rarer for unrelated meanings (for example, the meanings of ‘table’ and ‘finger’ are rather unlikely to be expressed by the same word in many languages)<sup>13,14,19</sup>. We provide a glossary of terminology in Supplementary Information (Supplementary Table 28).

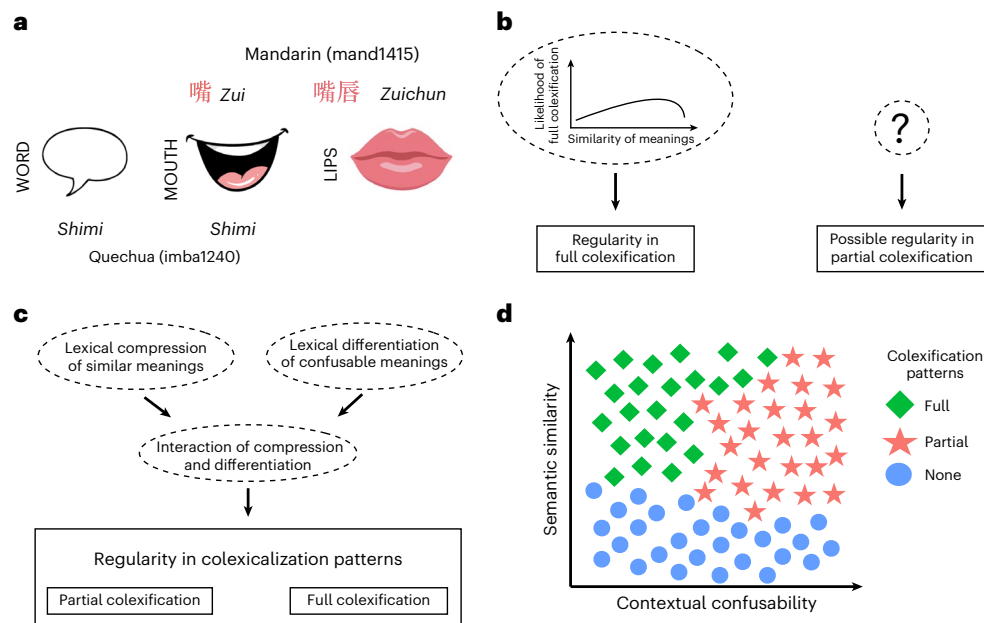
These cross-linguistic regularities refer to the mapping of meanings to entire word forms, regardless of whether they are simple or complex word forms. For example, *speak* in English can refer to the

action of using sounds to communicate (for example, *I speak with you*) or to the ability to talk in a language (for example, as in *I speak English*). In a similar way, *speaker*—which is composed of the base *speak* and the suffix *er*—can refer to a person who gives a speech, a person who knows a particular language or the piece of electrical equipment through which sound is emitted. In either case, two or more meanings are expressed by exactly the same form. These are cases of full colexification<sup>17</sup>. As illustrated in Fig. 1a, colexification is not unique to English (Indo-European). We find it across more and less distantly related languages: for example, in Spanish (Indo-European) the word *lengua* is used to refer to ‘tongue’ and ‘language’, and in Quechua (Quechuan), the word *shimi* is used to mean both ‘word’ and ‘mouth’.

However, languages do not only reuse entire word forms. At the subword level, lexical units are also systematically reused to express different meanings. For instance, English *grandfather* and *grandmother* share the prefix *grand*, which can be combined with a noun

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**Fig. 1 | Colexification patterns.** **a**, Examples of partial and full colexification. Quechua fully colexifies the meanings ‘word’ and ‘mouth’ in the form *shimi*. Mandarin partially colexifies ‘mouth’ and ‘lips’ in *zui(chun)*/嘴(唇). **b**, Pictorial summary of how regularity in full colexification is predicted by the similarity of meanings, monotonically increasing<sup>13</sup> until an inflection point is reached for highly similar meanings that appear in very similar contexts<sup>14,35</sup>.

By contrast, it is unclear whether partial colexification is regular and what would explain its regularity. **c**, Latent forces we hypothesize to lead to regularity in colexification, with an interaction of lexical compression and lexical differentiation explaining both partial and full colexification. **d**, Expected interaction of contextual confusability and semantic similarity (measures of the latent forces in **c**), yielding three distinct cross-linguistic lexicalization patterns.

to express one generation removed from it, such as the conventionalized *grand-aunt* or the less conventionalized *grand-supervisor*. Two meanings being mapped to partially but not fully overlapping forms is a case of partial colexification<sup>17,20,21</sup>. This phenomenon is particularly found in complex word forms—that is, morphologically complex word forms where at least two units are combined (for example, compounds or derived forms). As with full colexification, partial colexification is also pervasive across the languages of the world<sup>20,22</sup>. For example, as depicted in Fig. 1a, Mandarin Chinese (Sino-Tibetan), a language with very little subword morphology, uses the word *zui* (嘴) for ‘mouth’ and *zuichun* (嘴唇) for ‘lips’.

Little is yet known about the principles guiding partial colexification and its relationship to full colexification. The present study fills this gap by assessing whether there are cross-linguistic regularities in how meanings are mapped to forms at the subword level, how they compare to those at the word level and whether both are explainable through a set of general principles. In other words, we ask whether partial colexification is cross-linguistically regular, whether it shares commonalities with full colexification and whether they can be both explained in a unified way.

The pervasiveness of lexical ambiguity—stemming from full colexification—is explained by an array of functional advantages. It allows for more efficient signal compression and for the reuse of shorter and more preferred forms<sup>23–26</sup>. One efficient way of compressing the lexicon is to have semantically related meanings share the same form. Evidence from word learning studies suggests a bias towards forming categories and concepts composed of entities that share common properties and are therefore connected in conceptual space<sup>8,27–33</sup>. This bias facilitates the inductive task of concept formation from sparse input: because learners are never exposed to all possible contexts of use of a form, a prior preference for grouping similar entities helps constrain the space of candidate meanings. This learning bias also accounts for word meaning extension in language development and evolution<sup>34</sup> and for the resulting tendency of semantically similar meanings to be

expressed by the same form across the languages of the world<sup>13,14,35</sup>. In sum, fully colexifying related meanings facilitates lexical acquisition and retrieval<sup>11,12,15,19</sup>.

However, pressure to reduce the number and complexity of forms cannot operate in isolation; it must be balanced with pressure to accurately convey meaning in communication<sup>36,37</sup>. Too much of an overlap in the context in which meanings appear can counteract learning and processing advantages<sup>14,35</sup>: if two meanings are used in overlapping contexts, there is a tendency to map them to different forms. This is so because the functional advantages of full colexification can manifest only if meanings can be disambiguated in context<sup>24,26</sup>. Consequently, increased contextual similarity decreases full colexification likelihood. These findings are summarized in Fig. 1b, which depicts the forces shaping full colexification across languages.

We lack an analogous understanding of colexification below the word level. Our hypothesis is that partial colexification is the outcome of a balance between the same two forces that play a role in full colexification: lexical compression and differentiation. In particular, we hypothesize that partial colexification allows language users to still reap some benefits from reusing forms to express similar meanings while avoiding the meanings’ confusion in context by maintaining some difference in form. That is, we hypothesize that the lexical differentiation afforded by partial colexification is the next best lexicalization pattern available for cases where full colexification risks miscommunication. Under this view, partial colexification trades off lexical compression against lexical differentiation, being preferred midway between full colexification (same form, similar meanings) and full lexical differentiation (different form, dissimilar meanings). This hypothesis entails that we expect meanings that fully and partially colexify to differ. First, we expect low semantic similarity to be unlikely to lead to a recurrent co-expression of meanings. Second, if semantic similarity is high and contextual confusability is low, then languages will tend to use the same form for different meanings, making full colexification more likely. Third, as contextual confusability increases, so does the likelihood to lexically

differentiate meanings, with partial colexification becoming the preferred lexicalization pattern. The general idea of our unified hypothesis of lexicalization patterns at and below the word level is shown in Fig. 1c. Figure 1d sketches out how different degrees of semantic similarity and contextual confusability are predicted to interact by our hypothesis.

Although few studies exist on regularity in the cross-linguistic organization of meaning below the word level, there is support for the idea that both word and subword information are reused in efficient ways in the lexicon. Both full word reuse (full colexification) and word formation via processes such as derivation or compounding (partial colexification) have been found to be efficient solutions to lexicalize novel concepts as they emerge over time<sup>38</sup>. Although this account does not explain why certain concepts are encoded one way or the other, other research suggests that meanings involved in partial and full colexification may indeed differ. This evidence comes from the domain of perception verbs<sup>39</sup> and from two exploratory analyses<sup>20,22</sup>. Other lines of work have proposed that absolute usage frequency may influence the likelihood of full lexical differentiation as opposed to partial colexification, with more frequently used meanings being more likely to be fully differentiated<sup>21,40</sup>. However, these frequency effects alone cannot explain why two meanings might exhibit full versus partial colexification. We instead focus on confusability in context<sup>14,23,24</sup>, which more directly mediates the trade-off between learning and communication pressures that is involved in our hypothesis.

Our findings lend support to our idea of a gradient in cross-linguistic lexicalization patterns (Fig. 1c,d) explained by an interaction of universal pressure for the differentiation of confusable meanings and the compression of similar ones. In this way, our unified hypothesis enables a joint characterization of regularity at and below the word level across lexical domains.

## Results

### Lexicalization patterns differ in the meanings they recruit

We began by addressing the question of whether, cross-linguistically, form–meaning mappings differ across three kinds of lexicalization patterns (Fig. 1d): meanings mapped to different forms in a language (‘none’ pattern, with full lexical differentiation), those mapped to partially overlapping forms (partial colexification) and those mapped to the same form (full colexification). Establishing such differences was a necessary prerequisite for testing our main hypothesis. Specifically, we needed to observe systematic differences not only between cases of no colexification and colexification but also between different types of colexification. If the meanings recruited in each colexification type do not differ, this would suggest that all patterns reflect the same underlying principle, but there would be no reason to expect partial colexification to preferentially emerge in cases where full colexification would compromise successful information transfer.

We drew information about cross-linguistic form–meaning mappings from Lexibank<sup>41</sup>. This aggregated and curated resource has data on 1,995 languages from 193 language families. Partial colexifications are identified through partial matches in their form<sup>20</sup> (Supplementary Information). To decrease the number of false positives and make the analysis computationally tractable, we considered meanings to colexify only if they are identified to do so in at least four language families (see Supplementary Information for robustness checks showing qualitatively the same results if this threshold is changed to three or five). We did this to avoid classifying just any form overlap as partial colexification, focusing on cases with a clearer cross-linguistic signal. We obtained a dataset of 25,138 full colexifications and 193,587 partial colexifications across 1,982 languages and 192 families. In all analyses that follow, the number of meaning pairs that are lexically differentiated (none pattern) was subsampled to match the number of partial/full colexification pairs. See Supplementary Information for further details on pre-processing.

We compared the three patterns via the networks they give rise to<sup>42,43</sup>, with meanings as nodes and edges weighted by the number of

**Table 1 | Comparison of networks of meanings that cross-linguistically do not colexify, fully colexify or partially colexify, in terms of ARIs**

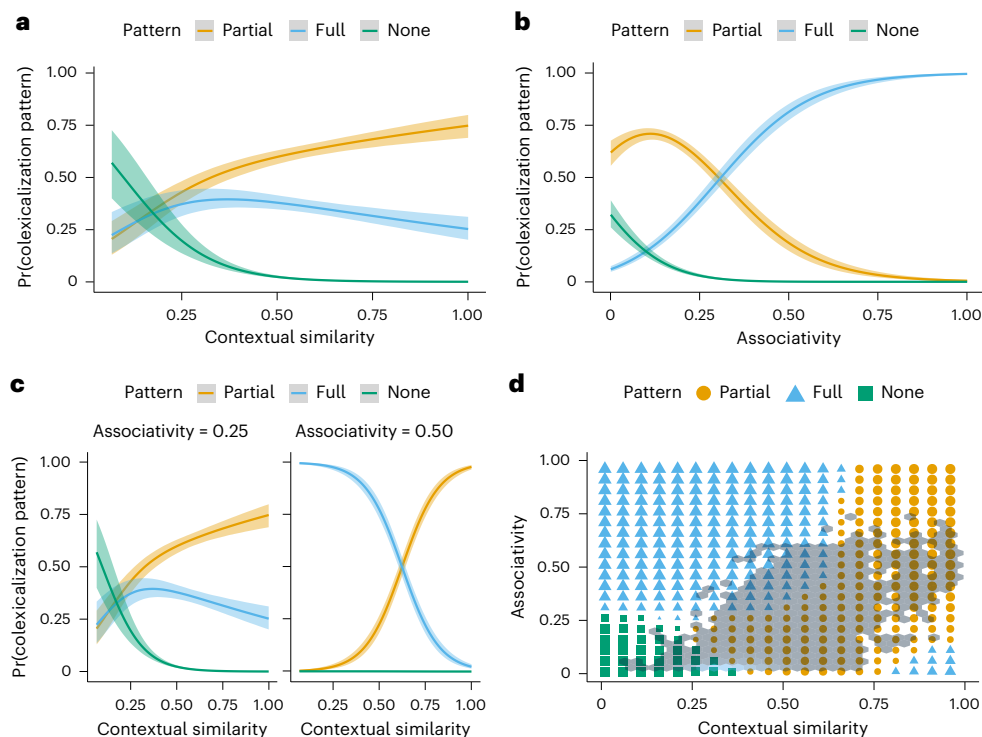
| (a)     | Imbalanced |         | Balanced |          |      |
|---------|------------|---------|----------|----------|------|
|         | Partial    | Full    | Partial  | Full     |      |
| None    | −0.0005    | 0.006   | 0.054    | 0.002    |      |
| Full    | 0.0626     |         | 0.183    |          |      |
| (b)     | Random     |         |          | Families |      |
|         | None       | Partial | Full     | Partial  | Full |
| None    | 0.001      |         |          |          |      |
| Partial | 0.85       |         | 0.35     |          |      |
| Full    | 0.81       |         |          | 0.66     |      |

A higher index indicates higher similarity. Comparisons across networks (a) quantify the similarity of lexicalization patterns to each other. ‘Imbalanced’ values compare all data available for each type of pattern, of differing sizes. ‘Balanced’ values are means of datasets of equal size. The bigger dataset was randomly downsampled 100 times to match the size of the smaller dataset in each comparison pair. Comparisons within networks of lexicalization patterns (b) indicate how similar the cross-linguistic distribution of meanings of each pattern is to itself. ‘Random’ values are means of 100 comparisons of random partitions of equal size of each dataset. ‘Families’ refers to means of pairwise comparisons of the networks of meanings that partially or fully colexify, split by language family. See Supplementary Information for details.

cases of no/partial/full colexification. To quantify the structural alignment between these patterns, we first identified discrete communities within each network using the InfoMap algorithm<sup>44</sup>. We then assessed the similarity of these community assignments using the adjusted Rand index (ARI) and adjusted mutual information (AMI). The ARI and AMI both speak to the similarity of networks. Here, this means that they quantify how similar the cross-linguistic distributions of the three lexicalization patterns are by evaluating their ‘closeness’: the ARI calculates the frequency with which pairs of nodes are grouped together (or kept separate), while AMI uses information theory to measure the reduction in uncertainty about one network’s structure given knowledge of the other. Both measures are adjusted for chance, meaning that a score of 0 represents the level of similarity expected from a random labelling of nodes, while 1 indicates a perfect topological match. Negative values indicate more frequent disagreement than what would be expected by chance. We report ARIs in the main text, but AMI yields consistent results (Supplementary Information).

Table 1a shows how the three patterns compare. The cross-linguistic distribution of meanings that partially and fully colexify is more similar (ARI of about 0.06, or 0.18 when balanced) than either is to the distribution of meanings that are not colexified (ARI of at least three magnitudes lower whether balanced or imbalanced). Table 1b quantifies within-pattern similarity across languages. The distribution of meanings that partially colexify is much more similar to itself (Table 1b)—both when split into random subsamples and when comparing across language families—than it is to full colexification (Table 1a). The same is true of full colexification. Moreover, our results suggest that the cross-linguistic distribution of partial colexifications is more heterogeneous than that of full colexifications (that is, a higher ARI in full than in partial), with partial colexifications differing more across families than full colexifications. A Bayesian beta regression predicting ARIs and AMI on the basis of pattern (partial versus full colexification) corroborates this ( $\beta_{\text{intercept}} = 0.7$  (0.6, 0.7),  $\beta_{\text{partial}} = -1.3$  (−1.4, −1.2),  $\phi_{\text{intercept}} = 3.1$  (2.9, 3.3),  $\phi_{\text{partial}} = -1$  (−1.2, −0.8), with 95% credible intervals; Supplementary Information).

In sum, the cross-linguistic distribution of partial and full colexifications is somewhat similar when contrasted against meanings that do not colexify. However, differences between partial and full colexification are also clearly suggested (compare refs. 20,22), not only by their modest ARIs (about 0.18) compared with within-pattern comparisons (0.35–0.85) but also by the higher heterogeneity of partial colexification across families.



**Fig. 2 | Estimates and predictions of the best model of cross-linguistic lexicalization patterns.** **a**, Marginal effect of contextual similarity. As contexts of use become more similar, fully colexifying becomes less likely than partially colexifying. **b**, Marginal effect of associativity. Closer associates are more likely to fully than to partially colexify. **c**, Interaction of context and associativity for two values of associativity (0.25 on the left, 0.50 on the right). As contextual similarity

increases, the likelihood of partially colexifying increases, with steeper slopes for closer associates. The shaded bands in **a–c** represent 95% credible intervals. **d**, Most likely lexicalization pattern based on the model posterior for possible combinations of associativity and contextual similarity values. Larger symbols indicate higher posterior probability. Dark hexagons mark regions of values attested in the data.

### Cross-linguistic lexicalization patterns are predicted by the interaction of similarity and confusability

To understand the differences identified above, we fit a series of Bayesian multinomial regressions. This analysis targets the two central tenets of our hypothesis: (1) that partial and full colexification can be explained by a common underlying principle—namely, the efficient trade-off between lexical compression and differentiation—and (2) that partial colexification tends to emerge in cases where full colexification may risk miscommunication. All models characterize colexification status—none/partial/full, with partial coded as the reference level—of meanings in a language. As predictors, we included proxies for contextual similarity and semantic similarity. As aforementioned, semantic similarity is linked to lexical compression (more similar meanings are more efficiently processed using shared forms), while contextual similarity allows us to assess the need for differentiation (meanings that appear in similar contexts should be expressed by different forms). Across five models, we used different combinations of these predictors: we evaluated an intercept-only model, as a baseline; two univariate models, each with only one proxy; one bivariate model; and a model with an interaction between contextual similarity and semantic similarity. All models estimate varying effects by language family for their population-level predictors. As a robustness check, all analyses were also conducted using binomial regressions focusing only on the contrast between partial and full colexification. These obtained very similar estimates and the same overall trends to those reported below (Supplementary Information).

As a proxy for semantic similarity, we used the English associativity data from the Small World of Words project<sup>45</sup>. Our target is a measure that provides a notion of how similar two meanings are in semantic memory. Associativity provides one way to operationalize this notion. Raw associativity data were obtained by asking human participants to

provide associates that came to their mind when prompted with a word. We transformed this data using a random-walk-based method that has been shown to provide the best fit to semantic similarity judgements<sup>45,46</sup> and be predictive of full colexification<sup>14</sup> as well as semantic change<sup>34</sup> (Supplementary Information). In contrast to contextual information—which also codifies information about the way meanings are (dis) similar to one another—associativity is not restricted to paradigmatic closeness. Although certain meanings that appear in similar contexts may also be close associates (for example, ‘dog’ and ‘cat’) and vice versa, this need not be the case. To illustrate, ‘bike’ and ‘car’ are distributionally similar in that they can be exchanged for one another in many contexts. However, they are not close associates. By contrast, ‘gas station’ is a close associate to ‘car’ but not to ‘bike’, and neither ‘car’ nor ‘bike’ is contextually similar to ‘gas station’<sup>47</sup>.

We used three English embedding models to quantify contextual similarity: fastText’s word-level and subword-level embeddings<sup>48</sup> and word2vec embeddings<sup>49</sup>. Intuitively, in this case our target measure is one that indicates the degree to which using the same form for two different meanings leads to ambiguity that may not be resolved in context. This would be the case if they often appear in the same linguistic environment. Embeddings offer a condensed representation of the context that a given word form appears in, fitting the above intuition. We report on the results with fastText’s word-level embeddings<sup>48</sup> in the main text since the estimates and overall trends do not vary much across the three embedding models (Supplementary Information). This suggests robustness with respect to this choice.

These operationalizations of semantic similarity and contextual similarity are proxies of the latent forces we aimed to model (Fig. 1c). In both cases, we chose to derive them from only English resources. This has the advantage that we could characterize the relationships of more meanings, as analogous large-scale resources do not exist for



other languages. The disadvantage is that English was used to quantify the relationships between meanings in other languages. We leave the adjudication of the degree to which this introduces a bias to future research for when better resources allow for a proper assessment (see also refs. 13,14,16,34 for cases where proxies for different languages align concerning broad cross-linguistic patterns).

We selected the best of the five models on the basis of how well it was estimated to predict out-of-sample data, using leave-one-out cross-validation<sup>50</sup>. Irrespective of the embedding model employed, the best model is the one that includes an interaction between associativity and contextual similarity, starkly outranking simpler models (Supplementary Information). Figure 2a shows the marginal effect of contextual similarity: as contextual similarity increases, partial colexification becomes more likely than full colexification, and not colexifying is less likely. In line with past research<sup>13,14,34</sup>, this model also predicts a positive effect of associativity on full colexification (Fig. 2b). Strong associates are more likely to fully colexify, with partial colexification becoming less likely. Meanings with low associativity are predicted to either partially colexify or not colexify at all. Figure 2c,d speaks more directly to our hypothesis. Figure 2c illustrates how associativity and context interact. In line with our expectations, the model suggests that, as contextual similarity increases, the likelihood that moderate to high associates are partially and not fully colexified increases as well. Finally, Fig. 2d gives a broader overview of the colexification pattern predicted to be the most likely, cross-linguistically, as a function of contextual similarity and associativity, synthesizing the interplay of the information depicted in Fig. 2a–c. These predictions closely align with our expectations (Fig. 1d).

To illustrate, our model predicts partial colexification to be the most likely lexicalization pattern for cases of high contextual similarity combined with medium associativity (for example, ‘long’–‘heavy’ or ‘fire’–‘ignite’) to high associativity (for example, ‘ten’–‘fourteen’, ‘hear’–‘listen’ or ‘ankle’–‘wrist’). The most likely cases of full colexification are found in the region of high associativity combined with low contextual similarity (for example, ‘breast’–‘suck’ or ‘foot’–‘footprint’) to medium contextual similarity (for example, ‘warm’–‘heat’ or ‘condemn’–‘convict’). Finally, meanings unlikely to colexify are those that are dissimilar in their context of use while not being close associates either (for example, ‘beard’–‘two’, ‘money’–‘meet’ or ‘heel’–‘parrot’).

## Discussion

The results of this study provide large-scale evidence for the existence of cross-linguistic regularities in partial colexification, and they shed light on the forces that shape these regularities. In particular, they suggest that partial colexification is a distinct yet systematic cross-linguistic strategy of form–meaning mapping, governed by an interplay of semantic similarity and contextual confusability. This complements the well-documented regularities in full colexification<sup>13,14,35</sup>, suggesting that the same latent forces—a drive towards lexical compression of similar meanings and one for lexical differentiation of confusable ones—operate across lexicalization patterns. Our findings thereby expand our understanding of universal principles at play in the organization of the world’s lexicons, explaining word and subword regularity through their interaction in a unified fashion. Specifically, partial colexification appears to arise as a middle-ground strategy when full colexification risks ambiguity in overlapping contexts.

While this explanation accounts for differences in the cross-linguistic distribution of the meanings involved in the three lexicalization patterns we studied (no colexification, partial colexification and full colexification), differences not explained by a trade-off between compression and differentiation remain. Our analyses suggest that partial colexifications are less homogeneous across language families than full colexifications. This may be due to typological variation in morphological productivity and phonological constraints, which we expected to interact with pressure for lexical compression and

differentiation. It may also point to the heterogeneous nature of processes that lead to partial colexification: different classes of meanings combine with different classes of morphological units by means of distinct processes.

In this study, cases of partial colexification include derived forms through affixation as well as compounding. These word-formation processes differ in their productivity within and across languages<sup>51</sup>. Future work should aim to analyse the pressures shaping each of these processes separately, taking into account the specific typologies of individual languages, which constrain both the types of morphological processes available and their degree of productivity. Moreover, we should extend the approach taken here to the inflected lexicon (that is, word forms including inflectional morphology) and to cases of partial colexification involving the reuse of non-morphemic units<sup>52–54</sup>—which can comprise cases such as those of thematic vowels or stem alternations in nominal declensions or verbal conjugations. Unlike most of the morphological material considered in this study, which is lexical and meaning-bearing, these inflectional morphemic and non-morphemic units encode grammatical meaning or do not encode semantic content at all. Although the productivity and reuse of inflectional and non-morphemic morphology follow efficient principles<sup>15,55</sup>, they do not necessarily obey the same pressures as shared lexical morphology. These matters highlight the need to also consider different phenomena separately in future work, examining the role of the different types of lexical and non-lexical morphology in enabling partial colexification across typologically diverse languages, and narrowing down on different morphological processes and the classes of meanings they involve.

As with partial colexification, full colexification also encompasses diverse phenomena such as polysemy, homonymy and underspecification<sup>17,18,21</sup>. While meaning associativity and contextual similarity provide good proxies for the trade-off between compressibility and confusability in cases of polysemy and underspecification, meaning associativity is not necessarily a reliable proxy for compressibility in homonymy. Homonymy instead probably reflects a general pressure for lexical compression, largely independent of more specific cognitive principles related to meaning representations. Two semantically distinct meanings expressed by the same word form are likely to be maintained as they allow for lexical compression while remaining fully disambiguated in context.

Domain-specific studies have already revealed additional layers of complexity to the organization of meaning<sup>39</sup> that our domain-general approach abstracts away from. In restricting our analysis to cases of colexification that are found in at least a few language families and to meanings for which English-based proxies exist, we have focused on a broad cross-linguistic signal. A consequence of these choices is that less well-represented or family-specific patterns fall out of the scope of our study (for example, the colexification of classifiers, which are comparatively infrequent in Lexibank). Further in-depth analysis of the way in which different domains conform to our findings and how this reflects back on the principles posited here remains to be explored.

More generally then, in this study, we have provided a cognitively grounded explanation for the cross-linguistic distribution of full and partial colexification in the lexicon. Further work assessing differences between distinct phenomena will deepen our understanding of variation in the strength of the compressibility–confusability trade-off across the lexicon.

## Methods

### Colexification data

We drew data on cross-linguistic lexicalization patterns from Lexibank<sup>41</sup>, a standardized meta-collection of wordlists. Lexibank provides phonetic transcriptions for forms (based on CLTS<sup>56,57</sup>) linked to unified identifiers for languages (from Glottolog<sup>58</sup>) and concepts (from Concepticon<sup>59</sup>). We applied a diversity threshold, retaining only

colexifications identified in at least four distinct language families. See Supplementary Section 3 for details. Supplementary Section 7 shows that our results also obtain for other more or less conservative diversity thresholds.

### Contextual similarity

Contextual similarity was measured through fastText's English word-level embeddings<sup>48</sup>, through their subword-level counterparts<sup>48</sup> or through word2vec embeddings<sup>49</sup>. The contextual similarity of two meanings, instantiated through the English word forms that express them, corresponds to the cosine similarity of their embeddings. See Supplementary Section 4 for details. Supplementary Section 7 shows that our results are robust across the three embedding models.

### Associativity

Associativity is used as a proxy for semantic similarity, based on the Small World of Words data<sup>45</sup>. Following previous research<sup>45,46</sup>, we transformed raw cue–response pairs from human participants, quantifying the associativity of cue  $c_i$  and cue  $c_j$ , based on their responses  $r$  as:

$$\text{asso}(c_i, c_j) = \frac{\sum_k G_{[r_k, c_i]} G_{[r_k, c_j]}}{\sqrt{\sum_k (G_{[r_k, c_i]})^2} \sqrt{\sum_k (G_{[r_k, c_j]})^2}}, \quad (1)$$

where

$$G = (I - \alpha P)^{-1}. \quad (2)$$

In equation (2),  $P$  is a Positive Pointwise Mutual Information-transformed and normalized cue–response matrix;  $I$  is the identity matrix and  $\alpha$  is a dampening parameter set to 0.75. See Supplementary Section 4.

### Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

### Data availability

All the data and resources that our analyses build on are freely available. In their original form, the data are available from the following sources. The Lexibank data<sup>41</sup> are available at <https://lexibank.cld.org/>. Raw human associativity data from the Small World of Words project<sup>45</sup> are available at <https://smallworldofwords.org/en/project/>. Word embeddings from fastText<sup>48</sup> are available at <https://fasttext.cc/> and those from word2vec at <https://code.google.com/archive/p/word2vec/>.

### Code availability

The data processing and analysis code used in this study is available via the Open Science Framework at <https://osf.io/qzaxk/>.

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## Author contributions

T.B. and C.S. conceptualized and designed the study. T.B., X.L. and J.D.W. acquired and analysed the data. T.B. and C.S. interpreted the analyses and wrote the paper. All authors approved the paper.

## Competing interests

The authors declare no competing interests.

## Additional information

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We analyze large scale cross-linguistic data, using network and regression analyses. This is a quantitative mixed-methods study. Data analyzed was collected in prior work.

Research sample

The sample is a convenience sample; collected, aggregated and normalized in previous research. It encompasses information about the way 3026 unique meanings are expressed, with data spanning across 1995 languages from 193 language families. See SI Section 3 for detailed discussion and reporting.

Sampling strategy

The data was collected in advance by other researchers. It is a convenience sample. Given the nature of the data, this is the state of the art in terms of a balance between breadth of diversity and quality. See SI Section 3 for detailed discussion and reporting. Both network analyses and regression analyses employ robustness checks to determine to which extent sampling biases may drive the results.

Data collection

n/a (see above)

Timing

n/a

Data exclusions

All data was used in network analyses. For regression analyses, data was excluded for those form-meaning pairs for which we lacked

|                   |                                                                                                                            |
|-------------------|----------------------------------------------------------------------------------------------------------------------------|
| Data exclusions   | predictor information (associativity and/or contextual similarity). See SI Section 7 for detailed discussion and reporting |
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